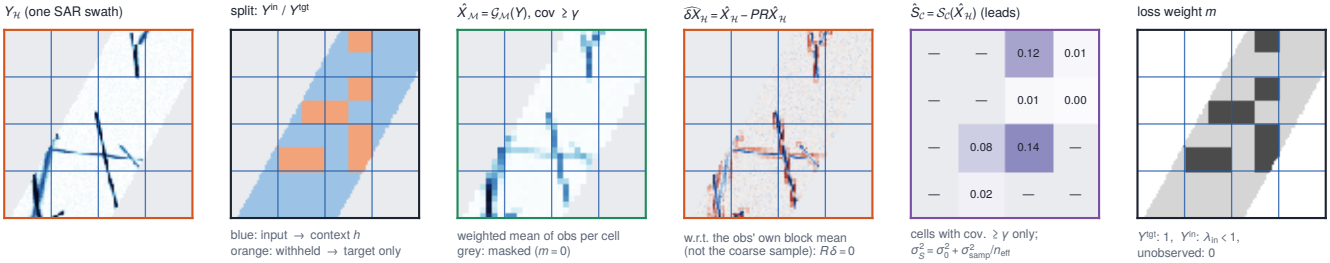
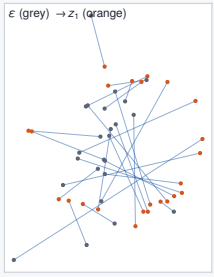
(a) One training batch, built on the fly: input / target split and ground truths



No precomputed truth: gridding $\mathcal{G}_\ell$, details, statistics $\mathcal{S}$, masks and the split are recomputed for every batch. Scores are always reported on Y^{tgt}.

(b) Conditional flow matching, one generator per level



$$z_s = s\,z_1 + (1-s)\,\epsilon, \quad u = z_1 - \epsilon$$

$$\hat{z}_1 = z_s + (1-s)\,v_\theta^\ell(z_s, s \mid \cdot)$$

ϵ projected by Π_0 for the details

$G_{\mathcal{C}}$: $Z_{\mathcal{C}} = (X_{\mathcal{C}}, S_{\mathcal{C}})$ | $h_{\mathcal{C}}, f$

$G_{\mathcal{M}}$: $Z_{\mathcal{M}} = (\delta X_{\mathcal{M}}, S_{\mathcal{M}})$ | $X^\eta_{\mathcal{C}}, \Pi^\downarrow \Phi_{\mathcal{C}}, h_{\mathcal{M}}, \Pi^\downarrow f$

$G_{\mathcal{H}}$: $\delta X_{\mathcal{H}}$ | $X^\eta_{\mathcal{C}}, \delta X^\eta_{\mathcal{M}}, \Pi^\downarrow \Phi_{\mathcal{M}}, h_{\mathcal{H}}, \Pi^\downarrow f$

training: X^η = sample of the level above (EMA model, no gradient) + noise η;
each context group dropped with prob. $p_\varnothing$ ⇒ heads $v(\cdot \mid h)$ and $v(\cdot \mid \varnothing)$. Inference: Heun, $\mathcal{C} \to \mathcal{M} \to \mathcal{H}$

(c) Losses (all masked, targets from (a))

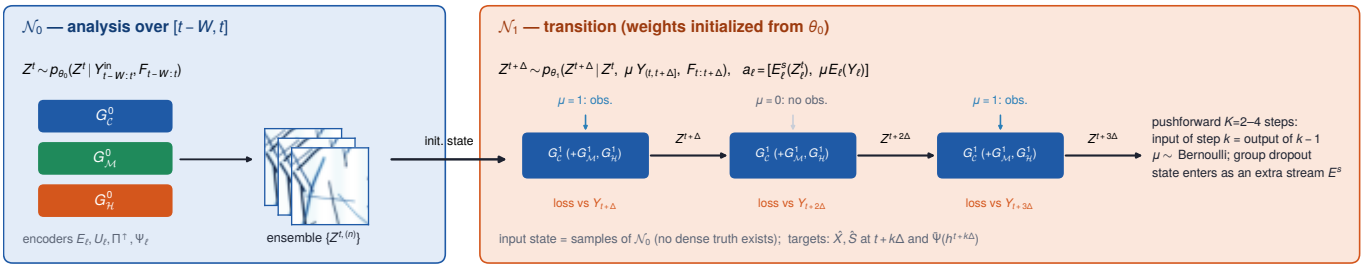
$$\mathcal{L}_{\text{NLL}}(u, v; \sigma, m) = \mathbb{E}\,\Sigma\, m \odot \left(\tfrac{1}{2}\left\|\tfrac{u-v}{\sigma}\right\|^2 + \log\sigma\right)$$

	term	target	mask
$\mathcal{L}_X^\ell$	state	$\hat{X}_\ell$ or $\widehat{\delta X}_\ell$	m_ℓ^X (var., level, pixel)
$\mathcal{L}_S^\ell$	explicit sub-grid	$\hat{S}_\ell = \mathcal{S}_\ell(\hat{X}_{\text{fine}})$	m^S, σ_S floor
$\mathcal{L}_{\text{cyc}}$	down/up consistency	$\|\mathcal{S}(\hat{x}_1^{\mathcal{H}}) - S_{\mathcal{C}}\|^2$	soft $\mathcal{S}$
$\mathcal{L}_\psi$	latent regularity	$\|\psi\|^2 + \lambda_\perp \|\text{Cov}(\psi, S)\|^2$	—
$\mathcal{L}_\psi^{\text{dyn}}$	latent in $\mathcal{N}_1$	$\bar{\Psi}(h^{t+\Delta})$ (EMA)	obs. at $t+\Delta$

$\mathcal{N}_0$: $\Sigma_\ell(\mathcal{L}_X^\ell + \lambda_S \mathcal{L}_S^\ell) + \lambda_c \mathcal{L}_{\text{cyc}} + \lambda_\psi \mathcal{L}_\psi$

$\mathcal{N}_1$: $\Sigma_{k=1}^K \gamma^{k-1} [\Sigma_\ell(\mathcal{L}_X^\ell + \lambda_S \mathcal{L}_S^\ell) + \lambda_\psi \mathcal{L}_\psi^{\text{dyn}}]_{t+k\Delta}$

(d) Two networks trained successively: initialization $\mathcal{N}_0$, then transition $\mathcal{N}_1$



Rollout, mode A: only $G^1_{\mathcal{C}}$ advances ($X_{\mathcal{C}}, S_{\mathcal{C}}, \psi_{\mathcal{C}}$); fine scales on demand with $G^0_{\mathcal{M}}, G^0_{\mathcal{H}}$ ($h = \varnothing$, conditioned by $\Pi^\downarrow \Phi$, $\Pi^\downarrow f$). Mode B (validation): full cascade at each step, then $S_{\mathcal{C}} \leftarrow \mathcal{S}_{\mathcal{C}}(X_{\mathcal{H}})$.